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Retrospective Evaluation of the Causes and Fluid Characteristics of Cavitary Effusions in Dogs and Cats

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ABSTRACT

Background: The relationship between fluid characteristics and cause of cavitary effusions is incompletely characterized.

Hypothesis/Objectives: Describe fluid characteristics and cytologic classification of cavitary effusions in dogs and cats.

Animals: A total of 269 dogs and 107 cats with cytologic fluid analysis of cavitary effusion.

Methods: Retrospective medical record review (2016–2020). Differences among groups were assessed using Kruskal–Wallis testing with post hoc pairwise Dunn's or Fisher's Exact testing.

Results: In dogs, peritoneal transudates most frequently were caused by decreased oncotic pressure (DOP; 63%, $p < 0.001$) and modified transudates by increased hydrostatic pressure (IHP; 41%, $p < 0.001$). Peritoneal exudates more frequently were caused by increased vascular permeability (IVP; 40%), ruptured viscera (35%), or neoplasia (23%) compared with IHP or DOP ($p < 0.01$ for all). Pleural transudates in dogs most frequently were caused by DOP (79%, $p < 0.001$) and modified transudates by IHP (33%) or neoplasia (29%). Pleural exudates were more likely to be caused by neoplasia (52%) or IVP (36%) compared with IHP or DOP ($p < 0.001$). In cats, peritoneal effusions commonly were caused by IVP or neoplasia, and pleural effusions by IHP or neoplasia. No statistical relationship was found between cytologic category and cause of effusion in cats.

Conclusions and Clinical Importance: Cytologic classification and fluid characteristics are most predictive of disease process in peritoneal effusions in dogs, whereas effusions in cats have substantial overlap across causes.

1 | Introduction

Pathologic cavitary effusions are defined as abnormal accumulations of fluid within the pleural, peritoneal, or pericardial spaces. Multiple disease processes can lead to fluid accumulation as a consequence of underlying pathophysiologic mechanisms, including abnormalities in intravascular hydrostatic pressure, plasma oncotic pressure, vascular permeability, or lymphatic drainage [1]. Recognizing the pathophysiologic mechanism causing cavitary effusion is required to formulate an appropriate diagnostic and treatment plan.

Clinicopathologic classification of cavitary effusion commonly is performed using cytologic fluid evaluation. Common categories of effusion based on nucleated cell count (NCC) and protein concentration include pure transudate (low protein and low cellularity), modified transudate (variable to moderate protein and cellularity), and exudate (high protein and high cellularity),

Abbreviations: CHF, congestive heart failure; DOP, decreased oncotic pressure; FIP, feline infectious peritonitis; IHP, increased hydrostatic pressure; IVP, increased vascular permeability; LMN, large mononuclear cell; NCC, nucleated cell count; N:M ratio, neutrophil to mononuclear cell ratio.

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but the specific criteria used to define these categories can vary [1–5]. Additional specific effusion characterizations also can be defined by cytologic features, including hemorrhagic, septic, and chylous effusions [3–5]. Cytologic evaluation also can be used to confirm the presence of an infectious agent or neoplastic disease [6]. In practice, this information is used to prioritize among differential diagnoses and direct further patient treatment decisions. In people, effusion analysis is used to determine prognosis and to identify patients who would benefit from more aggressive treatment interventions [7, 8].

Although fluid analysis often is used in the diagnostic process to suggest mechanisms of effusion in dogs and cats, no large-scale studies have investigated how well clinicopathologic classifications of effusion correlate with the underlying disease process. Previous studies of effusion characteristics in dogs have been limited by variable inclusion criteria and differences in definitions of effusion categories. Most published studies focus on a single disease process [9–13] or a single body cavity [6, 14–20]. Furthermore, many studies about effusion classifications do not report fluid and cytologic characteristics of the effusions [2, 3, 14, 17, 18].

Our primary objective was to describe cavitary effusion characteristics in a large population of dogs and cats with multiple underlying disease processes to determine which fluid characteristics would be most predictive of the pathophysiologic mechanism of the effusion. Secondary objectives included determining how effusion characteristics differ among cavities (peritoneal vs. pleural vs. pericardial) and between species (dog vs. cat) for a given mechanism of effusion.

2 | Materials and Methods

2.1 | Data Collection

A retrospective medical record search was performed to identify dogs and cats with documented cavitary effusion (pleural, peritoneal, or pericardial) with accompanying cytologic fluid analysis performed at Iowa State University Lloyd Veterinary Medical Center between March 21, 2016, and December 31, 2020. Cases were excluded if fluid analysis and cytology were not reported for the effusion or if the cause of the effusion could not be determined. Dogs and cats with multicavitary effusion were included if fluid analysis and cytology were performed on the effusion of at least one body cavity. Clinical data obtained from medical records included examination date, signalment, body weight, temperature, heart rate, respiratory rate, systolic blood pressure, and total number of cavities with effusion. Data collected for effusion fluid analysis included NCC, volume sampled, transparency, color, pH, protein concentration, packed cell volume, cytology findings, percentage of each cell type present in the effusion sample, neutrophil-to-mononuclear cell ratio (N:M ratio), relative number of red blood cells, and presence or absence of neutrophil degeneration, macrophage vacuolation, bacteria, neoplastic cells, and pathologic hemorrhage.

Effusions were classified as transudate, modified transudate, or exudate based on the relationship of protein concentration, NCC, and cell types present in cytologic analysis [21]. A transudate was defined as a protein concentration < 2.5 g/dL and < 1000 cells/μL with predominantly small and large mononuclear cells. A modified transudate was defined as a protein concentration > 2.5 g/dL and 1000–5000 cells/μL with predominantly small and large mononuclear cells and few neutrophils. An exudate was defined as a protein concentration > 2.5 g/dL and > 5000 cells/μL. The alternative specific effusion category of "hemorrhagic effusion" was defined as predominantly red blood cells (packed cell volume > 10%) with erythrophagocytosis, siderophages, or hematoidin crystals and protein concentration > 2.5 g/dL [2–4, 21].

When available, results of other diagnostic tests also were recorded, including CBC, serum biochemistry panel, other blood and urine testing, diagnostic imaging (abdominal radiographs, thoracic radiographs, computed tomography, echocardiography, abdominal ultrasonography), surgery or endoscopy reports, results of fine needle aspirates or surgical biopsies, and necropsy results.

Final diagnosis for the cause of each effusion was recorded based on information provided in the medical record. For dogs and cats for which the disease process causing effusion or the final diagnosis were unclear, additional investigators (J.L.W. and J.S.P.) performed independent assessments of these classifications, with final determination by consensus. Effusions were categorized based on primary mechanism as follows: increased vascular permeability (IVP, including inflammation), increased hydrostatic pressure (IHP), decreased oncotic pressure (DOP), neoplastic effusion, hemorrhagic effusion, or effusion secondary to ruptured viscera. For peritoneal effusions, the ruptured viscera category included rupture of the gastrointestinal tract, urinary bladder, gallbladder, or uterus. For pleural effusion, the ruptured viscera category included thoracic duct or lung lobe rupture. These mechanistic effusion categories were further subdivided based on final diagnosis for the underlying cause of the effusion.

2.2 | Statistical Analysis

Quantitative, continuous data was assessed for normality by Shapiro–Wilk analysis, with data presented as median (range)

TABLE 1 | Effusion location and number of cavities containing effusion in 269 dogs and 107 cats with cavitary effusion and cytologic analysis performed on at least one effusion.

	Dogs ($n = 269$)	Cats ($n = 107$)
Single cavitary	**208/269 (77%)**	**84/107 (78%)**
Peritoneal	155/208 (74%)	47/84 (56%)
Pleural	49/208 (24%)	37/84 (44%)
Pericardial	4/208 (2%)	0/84 (0%)
Bicavitary	**51/269 (19%)**	**18/107 (17%)**
Peritoneal + pleural	36/51 (71%)	11/18 (61%)
Peritoneal + pericardial	7/51 (14%)	2/18 (11%)
Pleural + pericardial	8/51 (15%)	5/18 (28%)
Tricavitary	**10/269 (4%)**	**5/107 (5%)**

TABLE 2 | Fluid and cytological characteristics of peritoneal effusion in 208 dogs separated by cause of peritoneal effusion. Results are provided as median (interquartile range) for continuous data, and as count (percent within column) for categorical data. Distinct numbers (n) are provided indicating all cases with the indicated effusion type (top row), cases with cytologic analysis of the effusion performed, and cases with sufficient information in cytologic analysis to allow classification of the effusion.

	All dog peritoneal effusion (*n* = 208)	**IVP (*n* = 55)**	**Neoplasia (*n* = 46)**	**IHP (*n* = 41)**	**Ruptured viscera (*n* = 37)**	**DOP (*n* = 27)**	**Hemorrhage (*n* = 2)**
Cases with fluid analysis performed (*n*)	**185**	**53**	**42**	**32**	**36**	**21**	**1**
NCC ($\times 10^3$/uL)	4280 (30–396 350)	8230 (100–396 350)	5495 (40–99 330)	1040 (80–8150)	22 810 (660–163 830)	280 (30–2400)	8680
pH	6.4 (6.0–7.2)	6.4 (6–7)	6.4 (6.4–7.2)	6.4 (6.2–7)	6.4 (6–7)	6.6 (6.4–7.2)	6.4
Protein (g/dL)	3.4 (0.1–6.6)	3.3 (0.5–6.6)	3.8 (0.7–6.4)	4.1 (0.2–6.0)	3.1 (0.3–5.7)	0.4 (0.1–2.8)	3.7
PCV (%)	11 (4.5–33)	6 (5–33)	15 (5–34)	NR	10 (6–11)	NR	22
Neutrophils (%)	80 (2–99)	87 (5–99)	67 (4–97)	67 (2–95)	95 (25–99)	77 (5–90)	97
Lymphocytes (%)	2 (0–94)	3 (0–27)	3 (0–94)	5 (0–16)	1 (0–20)	5 (1–30)	NR
Eosinophils (%)	0 (0–58)	0 (0–52)	0 (0–58)	0 (0–13)	0 (0–5)	0 (0–1)	2
MP or large MN cells (%)	11 (0–95)	10 (0–80)	9 (0–80)	24 (4–95)	0 (0–73)	20 (5–92)	1
N:M ratio	7.3 (0–99)	8.1 (0.0–99.0)	8.4 (0 0.2–97.0)	3.1 (0.0–22.0)	23.8 (0.3–99.0)	3.5 (0–18)	97
Cases with effusion classification evaluable (*n*)	**182**	**52**	**40**	**32**	**36**	**21**	**1**
Transudate	32 (18%)	1 (2%)	5 (13%)	4 (13%)	2 (6%)	20 (95%)	0 (0%)
Modified transudate	63 (35%)	17 (33%)	15 (37%)	26 (81%)	4 (11%)	1 (5%)	0 (0%)
Exudate	86 (47%)	34 (65%)	20 (50%)	2 (6%)	30 (83%)	0 (0%)	0 (0%)
Hemorrhagic	1 (0.5%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (100%)

Abbreviations: DOP, decreased oncotic pressure; IHP, increased hydrostatic pressure; IVP, increased vascular permeability; MN, mononuclear; MP, macrophage; NCC, nucleated cell count; N:M ratio, ratio of neutrophils to mononuclear cells; NR, not recorded; PCV, packed cell volume.

TABLE 3 | Fluid and cytological characteristics of pleural effusion in 103 dogs, separated by cause of pleural effusion. See Table 2 for the remainder of the key.

	All dog pleural effusion ($n=103$)	Neoplasia ($n=33$)	IVP ($n=25$)	IHP ($n=21$)	DOP ($n=13$)	Idiopathic chylothorax ($n=4$)	Hemorrhage ($n=4$)	Ruptured viscera ($n=3$)
Cases with fluid analysis performed	**85**	**31**	**21**	**11**	**13**	**4**	**2**	**3**
NCC ($\times 10^3$/uL)	5020 (40–122 030)	10 415 (110–101 030)	15 810 (660–122 030)	1470 (270–13 580)	245 (40–1570)	7465 (3690–39 230)	7100 (960–7520)	2890 (2330–5020)
pH	6.4 (6–7.6)	6.4 (6.4–7.2)	6.4 (6–7)	6.4 (6.4–7)	6.4 (6.4–7.6)	6.4 (6.4–6.4)	6.4 (6.4–6.4)	6.4 (6.4–6.8)
Protein (g/dL)	3.2 (0.2–6.3)	3.4 (0.2–5.2)	3.2 (1.8–5.0)	2.9 (1.8–4.6)	0.7 (0.0–5.4)	4.2 (3.4–6.3)	4.0 (3.9–4.9)	3.4 (2.7–4.7)
PCV (%)	5 (5–63)	33 (10–63)	NR	4.5	NR	NR	20 (14–26)	NR
Neutrophils (%)	41 (0–98)	35 (0–97)	87 (18–98)	40 (15–74)	25 (5–73)	14 (5–84)	29 (20–37)	15 (14–94)
Lymphocytes (%)	6 (0–100)	14 (0–100)	0 (0–47)	18 (0–56)	7 (0–35)	77 (1–88)	5 (4–5)	40 (1–71)
Eosinophils (%)	0 (0–50)	0 (0–50)	0 (0–6)	0 (0–4)	0 (0–11)	0 (0–0)	0 (0–0)	0 (0–0)
MP or large MN cells (%)	19 (0–89)	14 (0–72)	10 (0–65)	39 (11–72)	62 (12–89)	12 (4–15)	67 (58–76)	14 (5–46)
N:M ratio	2.0 (0–98.0)	2.0 (0.1–42.0)	8.0 (0.0–98.0)	0.7 (0.3–4.6)	0.5 (0.1–6.3)	2 (0.3–5)	0.5 (0.3–0.6)	1 (0.3–18.8)
Cases with effusion classification evaluable (n)	**82**	**31**	**18**	**11**	**13**	**4**	**2**	**3**
Transudate	14 (17%)	2 (6%)	0 (0%)	1 (9%)	11 (85%)	0 (0%)	0 (0%)	0 (0%)
Modified transudate	24 (29%)	7 (23%)	3 (17%)	8 (73%)	2 (15%)	2 (50%)	0 (0%)	2 (67%)
Exudate	42 (52%)	22 (71%)	15 (83%)	2 (18%)	0 (0%)	2 (50%)	0 (0%)	1 (33%)
Hemorrhagic	2 (2%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	2 (100%)	0 (0%)

Abbreviations: DOP, decreased oncotic pressure; IHP, increased hydrostatic pressure; IVP, increased vascular permeability; MN, mononuclear; MP, macrophage; NCC, nucleated cell count; N:M ratio, ratio of neutrophils to mononuclear cells; NR, not recorded; PCV, packed cell volume.

TABLE 4 | Fluid and cytological characteristics of pericardial effusion in 29 dogs, separated by cause of pericardial effusion. See Table 2 for the remainder of the key.

	All dog pericardial effusion ($n=29$)	Neoplasia ($n=19$)	Increased vascular permeability ($n=10$)
Cases with fluid analysis performed	**20**	**12**	**9**
NCC ($\times 10^3$/uL)	10 180 (350–46 960)	7830 (350–45 560)	23 360 (1190–46 960)
pH	6.4 (6.4–7.2)	6.4 (6.4–7.2)	6.4 (6.4–6.8)
Protein (g/dL)	4.1 (1.8–7.3)	4.5 (1.8–7.3)	3.2 (2.5–5.1)
PCV (%)	30 (10–58)	31 (10–58)	16 (14–37)
Neutrophils (%)	72 (14–95)	64 (14–95)	72 (14–95)
Lymphocytes (%)	4 (0–37)	10 (1–37)	3 (0–18)
Eosinophils (%)	0 (0–1)	0 (0–1)	0 (0–1)
MP or large MN cells (%)	25 (0–82)	26 (2–52)	25 (0–82)
N:M ratio	2.0 (0.2–43.0)	0.8 (0.3–43.0)	2.5 (0.2–8.0)
Cases with effusion classification evaluable (n)	**19**	**11**	**8**
Transudate	0 (0%)	0 (0%)	0 (0%)
Modified transudate	3 (16%)	2 (18%)	1 (12%)
Exudate	4 (21%)	1 (9%)	3 (38%)
Hemorrhagic	12 (63%)	8 (73%)	4 (50%)

or mean (95% confidence interval of mean) accordingly. Differences in continuous variables between categories were assessed either by Wilcoxon Rank Sum Test or by Kruskal–Wallis analysis with Dunn's correction for multiple comparisons. Categorical variables were compared between categories using Fisher's Exact Analysis. Statistical significance was defined as $p<0.05$ for all comparisons. Statistical analyses were performed using commercial software (GraphPad Prism v.10, GraphPad Software Inc.; MedCalc v.22, MedCalc Software Ltd).

3 | Results

3.1 | Case Demographics

A total of 431 cases were identified in the initial medical record search. Cases were excluded ($n=55$) if the underlying cause of effusion could not be determined based on the diagnostic testing performed ($n=46$) or if multiple different causes of effusion were clearly identified ($n=9$). A total of 376 cases met criteria for inclusion in the study, including 269 dogs and 107 cats. The median age for dogs was 8.4 years old (range, 1 month—16 years) and the median age for cats was 11 years old (range, 3 months—19 years). Among dogs, there were 145 males (119 neutered, 26 intact) and 124 females (106 spayed, 18 intact). Among cats, there were 71 males (68 neutered, three intact) and 36 females (33 spayed, three intact). The median weight for dogs was 21.5 kg (range, 2.0–81.3 kg) and for cats was 4.5 kg (range, 2.1–8.2 kg). The most commonly represented breeds of dogs were mixed breeds ($n=61$), Labrador retriever ($n=20$), golden retriever ($n=14$), Yorkshire terrier ($n=13$), and German shepherd ($n=10$). The most common breeds of cats were domestic shorthair ($n=68$), domestic longhair ($n=23$), Siamese ($n=4$), Maine Coon ($n=3$), and Russian blue ($n=2$).

3.2 | Disease Process Categories and Underlying Diseases Causing Cavitary Effusions

Of the 376 dogs and cats meeting inclusion criteria for the study, 292 had a single cavitary effusion, 69 had bicavitary effusion, and 15 had tricavitary effusion (see Table 1). Single cavitary effusion was found to be predominantly peritoneal in origin (202/292 [69%] cases of single cavitary effusion with species combined), whereas bicavitary effusions were most commonly peritoneal and pleural (47/69 [68%] cases of bicavitary effusion with species combined); tricavitary effusion was uncommon (15/376 [4%] total cases in both species). The cause of the effusion was considered definitive in 239/273 (88%) cases with peritoneal effusion, 131/161 (81%) cases with pleural effusion, and 30/41 (73%) cases with pericardial effusion. In other situations (75/475 [16%] total cavities), the cause of effusion was highly suspected based on clinical features including definitively diagnosed effusions in other body cavities. Effusion characteristics by species, cavity, and disease process category are reported in Tables 2–6.

For peritoneal effusions, the categories identified were IVP, neoplasia, IHP, ruptured viscera, DOP (dogs only), and hemorrhage (see Tables 2 and 5). In dogs, peritoneal effusion secondary to IVP was most commonly caused by peritonitis (50/55, 91%),

TABLE 5 | Fluid and cytological characteristics of peritoneal effusion in 65 cats, separated by cause of peritoneal effusion. See Table 2 for the remainder of the key.

	All cat peritoneal effusion ($n=65$)	Increased vascular permeability ($n=26$)	Neoplasia ($n=25$)	IHP ($n=7$)	Ruptured viscera ($n=5$)	Hemorrhage ($n=2$)
Cases with fluid analysis performed	**60**	**25**	**23**	**5**	**5**	**2**
NCC ($\times 10^3$/uL)	2105 (110–46 910)	1990 (370–32 680)	2810 (380–34 820)	880 (110–46 910)	3390 (880–40 130)	405 (150–660)
pH	6.4 (5.0–7.2)	6.4 (6.4–7.2)	6.4 (5.0–7.0)	6.4 (6.4–7.0)	6.6 (6.4–6.8)	6.4 (6.4–6.4)
Protein (g/dL)	3.5 (0.4–7.6)	4.0 (1.8–7.6)	3.5 (0.7–5.8)	4.1 (0.4–5.1)	2.0 (0.7–2.7)	2.0 (0.8–3.2)
PCV (%)	5 (4–13)	5 (4–5)	NR	NR	NR	13
Neutrophils (%)	73 (0–99)	76 (0–99)	63 (5–91)	40 (10–90)	95 (88–98)	78
Lymphocytes (%)	3.5 (0–90)	2.3 (0–41)	13 (0–90)	4 (0–40)	0 (0–1)	12
Eosinophils (%)	0 (0–5)	0 (0–0)	0 (0–5)	0 (0–0)	0 (0–1)	0
MP or large MN cells (%)	15 (0–95)	15 (1–95)	16 (0.3–69)	29 (10–86)	5 (2–10)	10
N:M ratio	4 (0–99)	5.3 (0–99)	2.9 (0–91)	2 (0–9)	19 (8–49)	7
Cases with effusion classification evaluable (n)	**54**	**22**	**23**	**3**	**5**	**1**
Transudate	5 (9%)	2 (9%)	1 (4%)	1 (33%)	1 (20%)	0 (0%)
Modified transudate	31 (57%)	14 (64%)	14 (61%)	1 (33%)	2 (40%)	0 (0%)
Exudate	17 (32%)	6 (27%)	8 (35%)	1 (33%)	2 (40%)	0 (0%)
Hemorrhagic	1 (2%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (100%)

TABLE 6 | Fluid and cytological characteristics of pleural effusion in 58 cats, separated by cause of pleural effusion. See Table 2 for the remainder of the key.

	All cat pleural effusion ($n=58$)	IHP ($n=22$)	Neoplasia ($n=19$)	Increased vascular permeability ($n=8$)	Idiopathic chylothorax ($n=7$)	DOP ($n=1$)	Hemorrhage ($n=1$)
Cases with fluid analysis performed	**50**	**20**	**15**	**7**	**7**	**1**	**0**
NCC ($\times 10^3$/uL)	5040 (70–68 820)	3025 (70–37 030)	11 780 (440–17 710)	3895 (1190–68 820)	6970 (4260–38 040)	1370	NR
pH	6.4 (6.0–7.0)	6.4 (6.2–7.0)	6.4 (6.0–6.8)	6.4 (6.2–6.8)	6.4 (6.4–6.8)	7.0	NR
Protein (g/dL)	3.9 (0.1–8.0)	3.1 (1.9–6.0)	4.2 (3.1–5.9)	4.5 (0.1–8.0)	4.3 (3.9–5.4)	1.1	NR
PCV (%)	NR	NR	NR	NR	NR	0	NR
Neutrophils (%)	22 (1–95)	18 (1–72)	23 (1–87)	77 (2–95)	22 (7–38)	85	NR
Lymphocytes (%)	15 (3–80)	40 (3–80)	27 (7–50)	11 (5–17)	10 (5–10)	0	NR
Eosinophils (%)	40 (0–97)	45 (5–95)	54 (0–97)	5 (0–83)	70 (59–88)	5	NR
MP or large MN cells (%)	0 (0–5)	1 (0–1)	0 (0–0)	0 (0–0)	2.5 (1–5)	0	NR
N:M ratio	10 (0–87)	15 (0–87)	27 (1–76)	16 (5–56)	3 (0.7–10)	0	NR
Neutrophils (%)	2.3 (0.0–54.3)	1 (0.0–17.0)	1.3 (0.0–6.7)	3.8 (0.1–19.0)	5.5 (2.0–54.0)	8.0	NR
Cases with effusion type evaluable	**49**	**20**	**15**	**6**	**7**	**1**	**0**
Transudate	4 (8%)	3 (15%)	0 (0%)	1 (17%)	0 (0%)	0 (0%)	NR
Modified transudate	19 (39%)	10 (5%)	5 (33%)	2 (33%)	1 (14%)	1 (100%)	NR
Exudate	26 (53%)	7 (35%)	10 (67%)	3 (50%)	6 (86%)	0 (0%)	NR
Hemorrhagic	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	NR

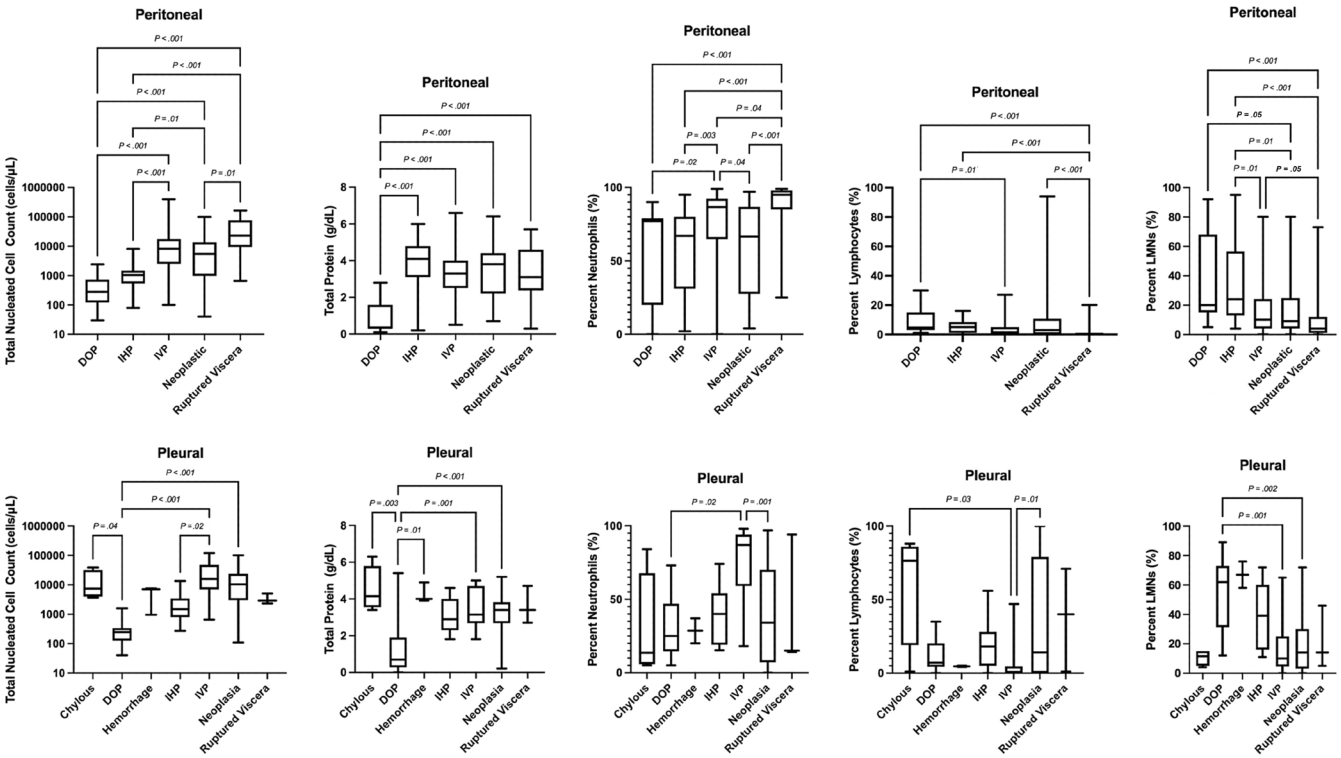

FIGURE 1 | Comparison of effusion characteristics among disease process categories within specific body cavities for dogs. For all box and whisker plots, the median is represented by a horizontal line, boxes indicate interquartile range, and whiskers indicate range. DOP, decreased oncotic pressure; IHP, increased hydrostatic pressure; IVP, increased vascular permeability; LMN, large mononuclear cells.

whereas in cats, an inflammatory peritoneal effusion usually was associated with either feline infectious peritonitis (FIP; 12/26, 46%) or other causes of peritonitis (13/26, 50%). For dogs with neoplastic peritoneal effusion, lymphoma (12/46, 26%) and hemangiosarcoma (9/46, 20%) were most frequently identified. For cats, neoplastic peritoneal effusions most often were caused by lymphoma (9/25, 36%) or pancreatic adenocarcinoma (6/25, 24%). For dogs with peritoneal effusion secondary to IHP, right-sided congestive heart failure (CHF; 23/41, 56%) and portal hypertension secondary to hepatic disease (6/41, 15%) were the most common underlying diseases. In cats, peritoneal effusion from IHP was associated primarily with CHF (3/7, 43%). Peritoneal effusion secondary to ruptured viscera most often was associated with septic peritonitis (30/37, 82%) or uroabdomen (6/37, 16%) in dogs, whereas for cats it was most often associated with uroabdomen (3/5, 60%). For dogs with peritoneal effusion secondary to DOP, hypoalbuminemia was most commonly secondary to protein-losing enteropathy (11/27; 41%) or end-stage liver failure (7/27, 26%).

For pleural effusions, disease process categories were neoplasia, IVP, IHP, DOP, idiopathic chylothorax, hemorrhage, and ruptured viscera (dogs only; Tables 3 and 6). In dogs, neoplastic pleural effusion was most commonly associated with lymphoma (10/33, 30%) or pulmonary adenocarcinoma (7/33, 21%), whereas in cats, lymphoma (6/19, 32%) was most common. For dogs with pleural effusion secondary to IVP, a septic or systemic inflammatory process most often was identified with pyothorax (13/25, 52%) and pleuritis (4/25, 16%) reported most commonly. For cats, the most common cause of pleural effusion secondary to IVP was pyothorax (4/8, 50%). In dogs, the most common causes of pleural effusion secondary to IHP were right-sided CHF (8/21, 38%) and lung lobe torsion (2/21, 10%). In cats, pleural effusion secondary to IHP was caused by CHF in all (22/22, 100%) cases. For dogs with pleural effusion secondary to DOP, hypoalbuminemia was most commonly secondary to protein-losing enteropathy (10/13, 76%) or end-stage liver failure (2/13, 15%). Pleural effusions classified as chylothorax were deemed idiopathic in all dogs (4/4, 100%) and cats (7/7, 100%). For dogs with pleural effusion secondary to hemorrhage, a coagulopathy often was involved (2/4, 50%). For dogs with pleural effusion secondary to ruptured viscera, rupture of the thoracic duct from trauma (2/3, 66%) was most frequent.

For pericardial effusions in dogs, the categories were neoplasia and IVP (see Table 4). Neoplasia was most frequent (19/29, 66%) with hemangiosarcoma being the most common type of neoplasia (12/19, 63%). Idiopathic pericarditis (10/29, 34%) was the second most common cause of pericardial effusion in dogs. For the 12 cases of pericardial effusion in cats, cytologic evaluation was not performed on any pericardial effusion samples. However, based on other diagnostic test results and characteristics of effusion in other body cavities, causes of pericardial effusion in cats were IHP (7/12, 58%; all with CHF), neoplasia (4/12,33%), and IVP (1/12, 8%; FIP).

3.3 | Comparison of Effusion Characteristics Among Disease Process Categories Within a Given Species and Body Cavity

In dogs, several clinicopathologic characteristics differed significantly among categories for peritoneal and pleural effusions (see

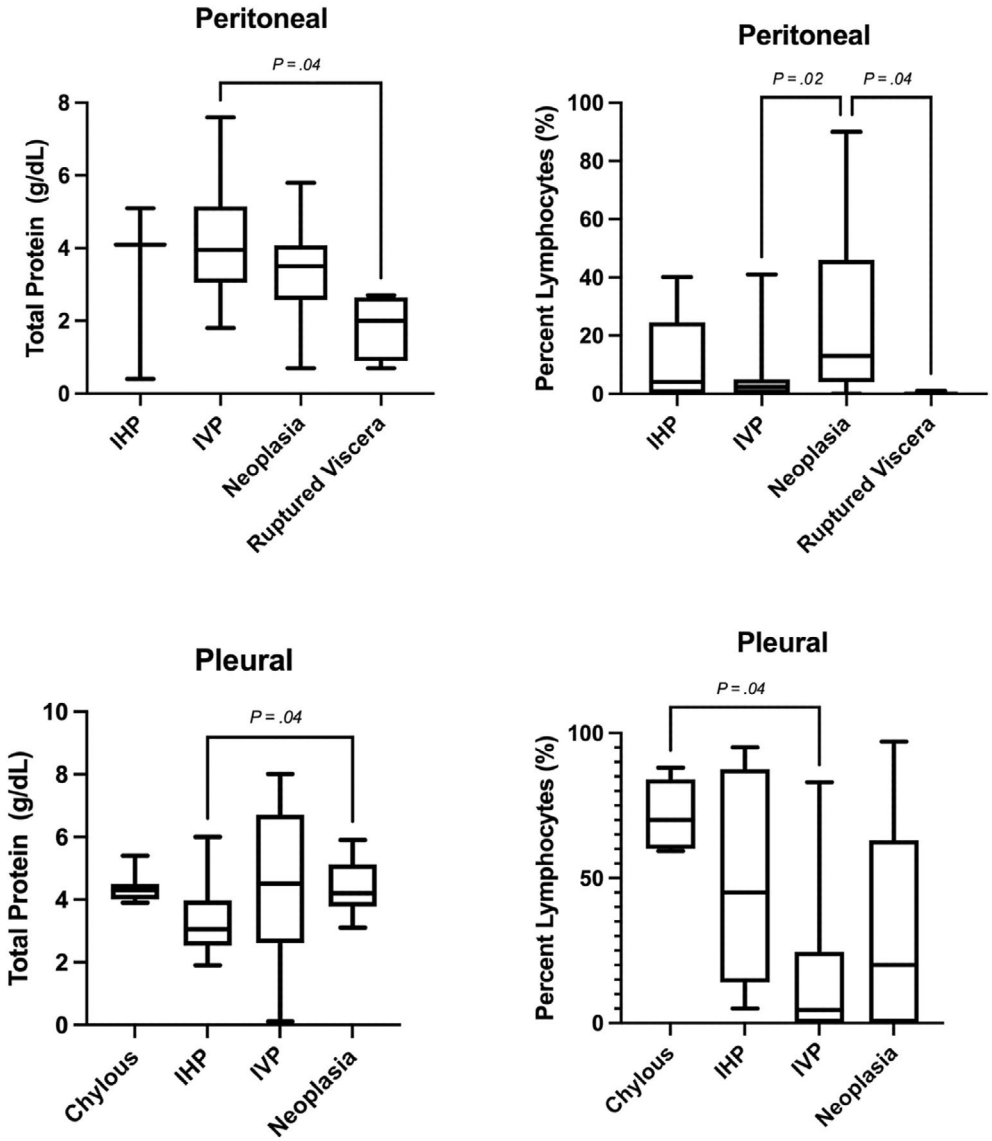

FIGURE 2 | Comparison of effusion characteristics among disease process categories within specific body cavities for cats. For all box and whisker plots, the median is represented by a horizontal line, boxes indicate interquartile range, and whiskers indicate range. IHP, increased hydrostatic pressure; IVP, increased vascular permeability.

Figure 1 and Table S1). For peritoneal effusions in dogs, those caused by DOP had lower NCC and total protein concentration compared with other causes, whereas effusions caused by IHP had lower NCC compared with other causes. The percentage of neutrophils was highest in ruptured viscera and IVP compared with other causes. Degenerate neutrophils also were observed more frequently in ruptured viscera (69%) compared with IVP (40%, $p = 0.01$), DOP (37%, $p = 0.04$), IHP (13%, $p < 0.001$), and neoplasia (13%, $p < 0.001$). For pleural effusions in dogs (see Table S1), effusions caused by DOP had lower NCC, lower total protein concentration, and higher percentage of mononuclear cells compared with other causes. Inflammatory effusions had higher percentages of neutrophils and higher N:M ratios compared with other causes, as well as higher NCC compared with IHP effusion. Additionally, degenerate neutrophils were observed more frequently in inflammatory effusions (75%) as compared with effusions that were chylous (0%, $p = 0.03$), caused by IHP (14%, $p = 0.01$), ruptured viscera (0%, $p = 0.03$), or neoplasia (17%, $p = 0.003$).

In cats, fewer differences in fluid characteristics were found among disease process categories. Peritoneal effusions caused by IVP had higher total protein concentration than those caused by ruptured viscera ($p = 0.04$; Figure 2). Neoplastic peritoneal effusions had a higher percentage of lymphocytes compared with effusions caused by IVP or ruptured viscera ($p = 0.02$ and $p = 0.04$, respectively). Pleural effusions in cats caused by IHP had a lower total protein concentration than effusions caused by neoplasia ($p = 0.04$). Additionally, chylous pleural effusions in cats had a higher percentage of eosinophils compared with those caused by IHP, inflammatory, or neoplastic pleural effusions ($p < 0.001$), and chylous effusions had a higher percentage of lymphocytes compared with inflammatory effusions ($p = 0.04$).

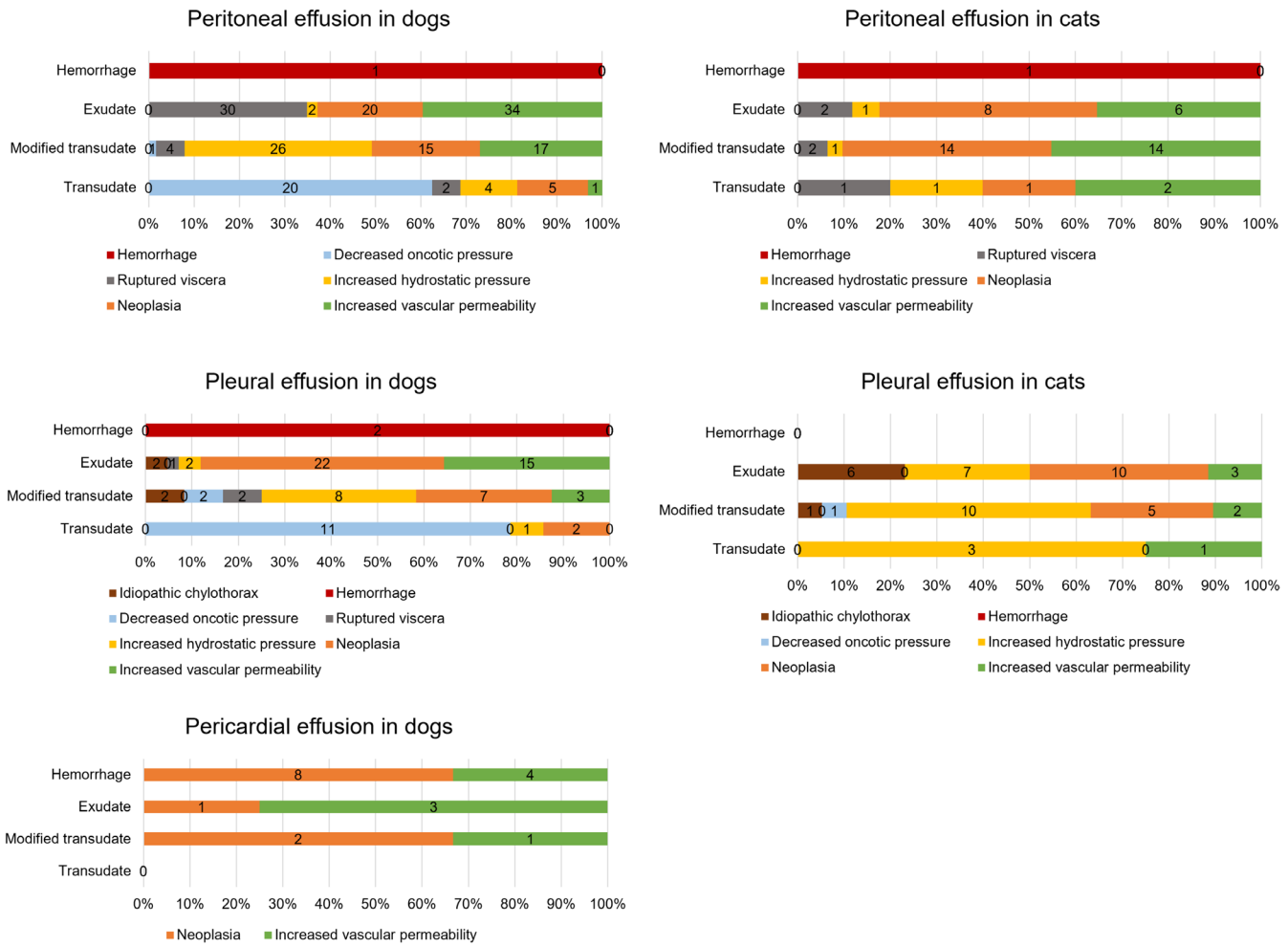

FIGURE 3 | Percentage of cytologic classifications (transudate, modified transudate, exudate, or hemorrhage) caused by various underlying causes of effusion in dogs and cats. Numbers within each bar indicate absolute numbers of animals with each cytologic classification for each cause; see Tables 2–6 for detailed breakdown.

No significant differences were found among disease process categories for pericardial effusion in either dogs or cats.

3.4 | Relationship Between Cause of Effusion and Cytologic Classification

The relationship between cytologic classification (transudate, modified transudate, or exudate) and the underlying disease process causing the effusion is presented in Figures 3 and S1, with details (counts and percentages) provided in Tables 2–6.

For peritoneal effusions in dogs, effusions classified as transudates were most likely to be caused by DOP (20/32 [63%] transudates; $p < 0.001$ compared with all other causes). Peritoneal modified transudates most commonly were caused by IHP (26/63 [41%]), significantly more commonly than any other cause ($p < 0.001$). Modified transudates also were commonly caused by IVP (17/63 [27%]; more common than ruptured viscera [$p = 0.02$] or DOP [$p = 0.02$]) and by neoplasia (15/63 [24%]; more common than ruptured viscera [$p = 0.01$] or DOP [$p = 0.01$]). Peritoneal exudates were caused by IVP (34/86 [40%]; more common than IHP or DOP [$p < 0.001$ for both]); ruptured viscera (30/86 [35%]; more common than neoplasia, IHP, or DOP [$p < 0.01$, $p = 0.001$, and $p = 0.001$, respectively]); or neoplasia (20/86 [23%]; more common than IHP or DOP [$p < 0.001$ for both]).

For pleural effusions in dogs, effusions classified as transudates were most likely to be caused by DOP (11/14 [79%]; $p = 0.03$ compared with ruptured viscera, $p = 0.004$ compared with chylous, and $p < 0.001$ for all other causes). Pleural modified transudates in dogs most commonly were caused by IHP or neoplasia (8/24 [33%] and 7/24 [29%], respectively), with a significantly higher proportion observed for IHP relative to IVP ($p = 0.01$) or DOP ($p = 0.02$). Exudative pleural effusions in dogs most commonly were caused by neoplasia (22/42 [52%]; more common than IHP [$p = 0.03$] and DOP [$p < 0.001$]), with the second most common cause being IVP (15/42 [36%]; more common than IHP [$p < 0.001$] or DOP [$p < 0.001$]). Pericardial effusions in dogs were classified most commonly as hemorrhagic (12/19, 63%) with no relationship between cytologic classification and underlying disease process (neoplasia or IVP).

For peritoneal effusion in cats, the most common causes were IVP and neoplasia, and no statistical relationship was found between cytologic classification and cause of effusion. For pleural effusion in cats, the most common causes were IHP and

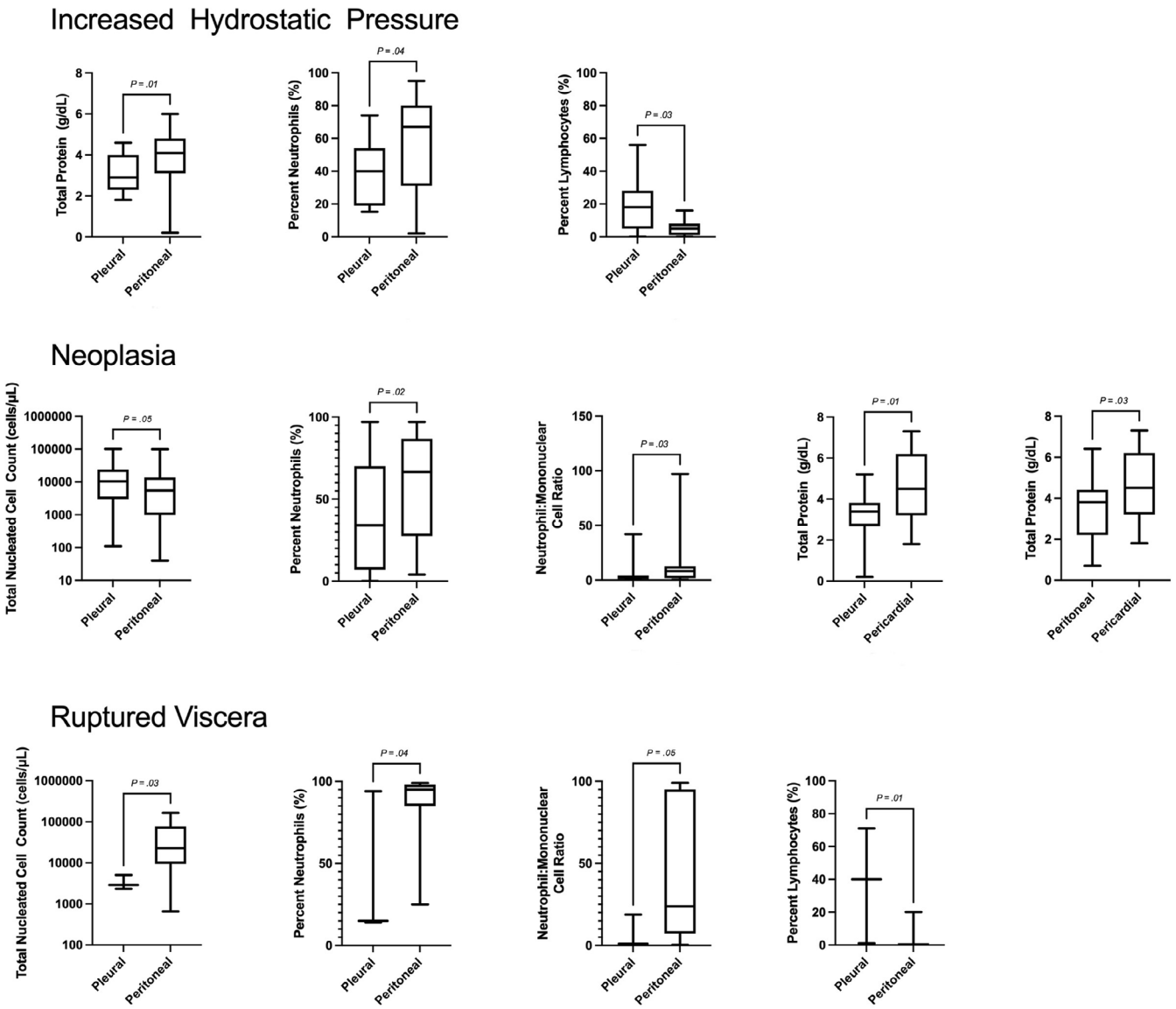

FIGURE 4 | Comparison of effusion characteristics between pleural and peritoneal cavities within a specific disease process category in dogs. For all box and whisker plots, the median is represented by a horizontal line, boxes indicate interquartile range, and whiskers indicate range.

neoplasia. Although all neoplastic effusions were either modified transudates or exudates, no significant relationship was found between cytologic category and cause of effusion.

3.5 | Comparison of Effusion Characteristics Among Body Cavities

In dogs, particular disease processes led to different fluid characteristics based on the cavity involved (Figure 4). For IHP, peritoneal effusions had higher total protein concentration ($p = 0.01$), higher percentage of neutrophils ($p = 0.04$), and lower percentage of lymphocytes ($p = 0.03$) compared with pleural effusions. Neoplastic effusions were more likely to be hemorrhagic when they occurred in the pericardium compared with pleural ($p < 0.001$) or peritoneal ($p < 0.001$) cavities. Neoplastic effusions in the peritoneal cavity had lower NCC ($p = 0.05$), higher percentage of neutrophils ($p = 0.01$), higher N:M ratio ($p = 0.03$), and were less likely to be classified as exudates ($p = 0.004$) compared with neoplastic effusions in the pleural cavity. Neoplastic pericardial effusions also had higher total protein concentrations compared with neoplastic pleural ($p = 0.01$) or peritoneal ($p = 0.03$) effusions. Peritoneal effusions secondary to ruptured viscera had higher NCC ($p = 0.03$), higher percentage of neutrophils ($p = 0.04$) and higher N:M ratio ($p = 0.05$), and lower percentage of lymphocytes ($p = 0.01$) compared with ruptured viscera in the pleural space.

For cats, the comparison among cavities was more limited because of a lower number of cases in certain disease processes, but some significant differences between cavities still were noted (see Figure 5). For effusions secondary to IHP, peritoneal effusions had a lower percentage of lymphocytes ($p = 0.01$) compared with pleural effusions. Neoplastic effusions in the peritoneal cavity had lower total protein concentrations ($p = 0.01$), higher percentages of neutrophils ($p = 0.01$), and higher N:M ratios ($p = 0.01$) compared with neoplastic effusions in the pleural space.

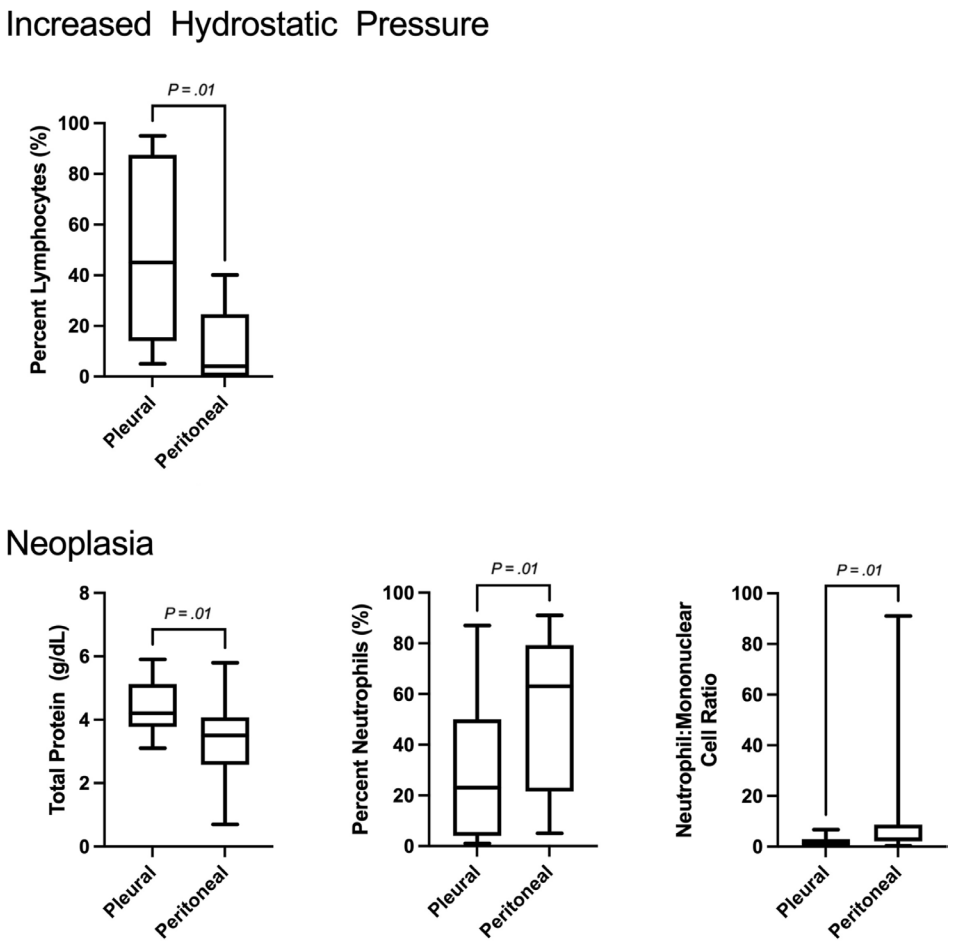

FIGURE 5 | Comparison of effusion characteristics between pleural and peritoneal cavities within a specific disease process category in cats. For all box and whisker plots, the median is represented by a horizontal line, boxes indicate interquartile range, and whiskers indicate range.

3.6 | Comparison of Effusion Characteristics Between Species

Effusions from dogs differed from those of cats in several ways when considering a single body cavity and cause of effusion (Figure 6). For peritoneal effusions secondary to IVP, effusions in dogs had higher NCC ($p = 0.001$) compared with cats and were more likely to be categorized as exudates ($p = 0.01$). For neoplastic peritoneal effusion, effusions in cats had higher percentages of lymphocytes compared with dogs ($p = 0.02$). For neoplastic pleural effusion, effusions in cats had higher total protein concentrations compared with dogs ($p < 0.001$). Chylous pleural effusions had higher percentage of eosinophils in cats compared with dogs ($p = 0.01$). No other significant differences were found between species for individual cavities and causes of effusion.

4 | Discussion

Veterinary textbooks frequently utilize NCC and protein concentration of cavitary effusions to apply the classifications of transudate, modified transudate, and exudate [1, 2, 5]. Classically, transudates are thought to be associated with the mechanism of DOP or IHP, exudates are associated with inflammatory or septic processes, and modified transudates are associated with IHP or long-term presence of transudates [1, 2, 21, 22]. Within the category of IHP, CHF and postsinusoidal portal hypertension are reportedly more likely to cause modified (or high-protein) transudates, whereas lymphatic obstruction or presinusoidal or sinusoidal portal hypertension may be more likely to cause low-protein transudates [1]. Overall, results of our study suggest that fluid classification (transudate, modified transudate, or exudate) might have the most diagnostic value for peritoneal effusion in dogs, whereas considerable variation in fluid characteristics was found in other cavities and species. Differential diagnoses for cavitary effusions in dogs and cats should not be ruled out on the basis of cytologic fluid classification alone.

Data from our study regarding the distribution of effusions among cavities is challenging to compare to previous literature, which typically focused on only a single body cavity [10, 15, 16] or combined all cavities together for analysis [3]. In one study of 81 dogs with pleural or mediastinal effusion, 31% had concurrent peritoneal effusion [14], and in 527 dogs with peripheral edema, 29% had concurrent abdominal effusion [23], supporting the notion that peritoneal effusion is the predominant cavitary effusion in dogs.

In dogs in our study, the most common causes of peritoneal effusion were IVP, neoplasia, IHP, and ruptured viscera. This finding is similar to previous literature reporting the most common causes of peritoneal effusion in dogs being either

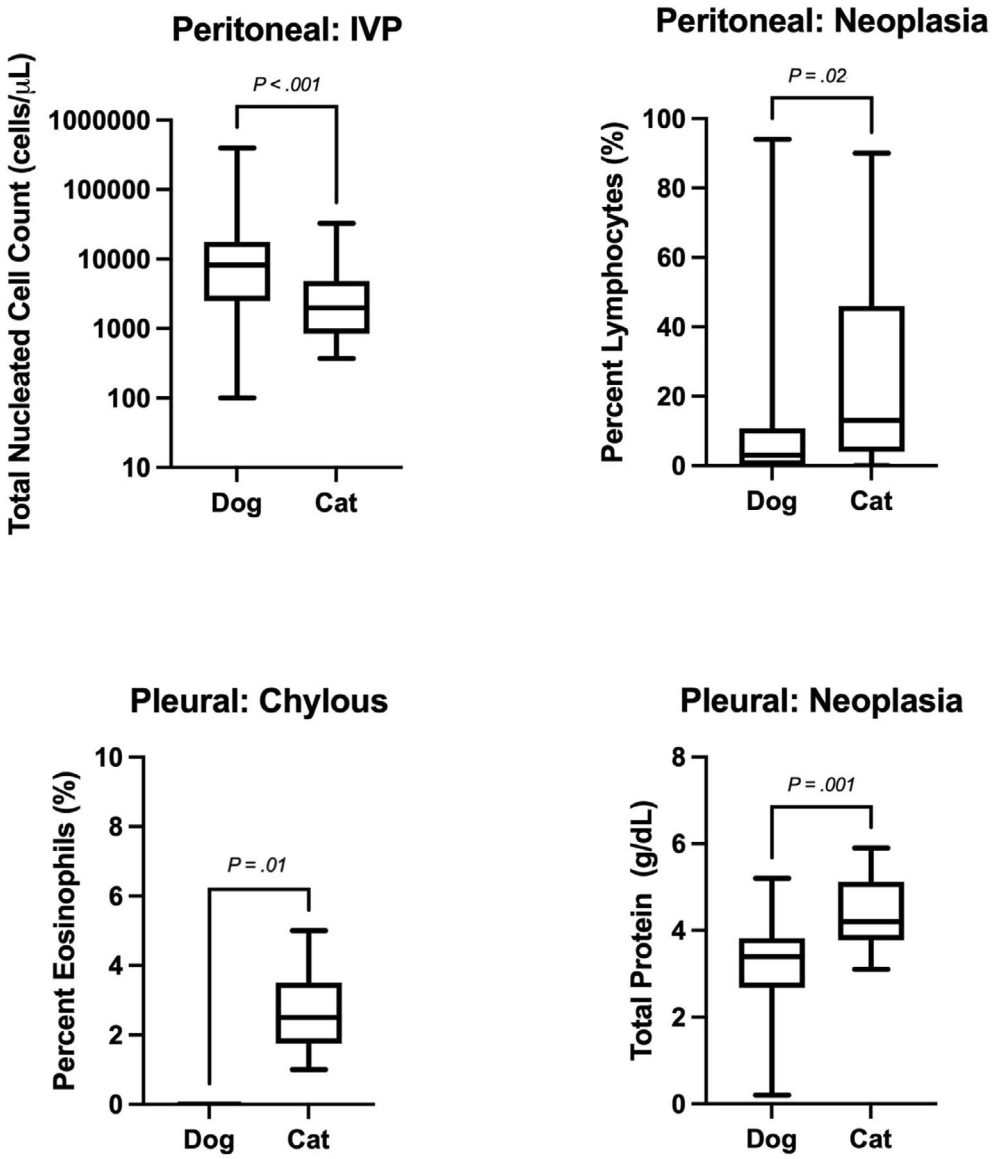

FIGURE 6 | Comparison of effusion characteristics between dogs and cats for a given body cavity and disease process category. For all box and whisker plots, the median is represented by a horizontal line, boxes indicate interquartile range, and whiskers indicate range.

CHF, neoplasia, and liver disease [4] or CHF, neoplasia, and septic peritonitis [13]. Peritoneal effusions caused by ruptured viscera frequently were classified as exudates (83%) and had higher degenerate neutrophil counts than other causes. This observation is consistent with the expected pathophysiology of ruptured abdominal viscera leading to increased endothelial permeability, release of inflammatory mediators, and eventual sepsis [2, 12]. Peritoneal effusions caused by IVP were classified as exudates (65%) or modified transudates (33%), and had a higher frequency of degenerate neutrophils compared with other causes except for ruptured viscera, again consistent with the expected pathophysiology. Although not among the most common causes of peritoneal effusion in dogs, DOP differed from other causes in having lower total protein concentration, NCC, and pH, and being more likely classified as transudates. Lower total protein concentration and transudative effusion are consistent with the expected pathophysiology of hypoalbuminemia and have been previously reported [3, 11]. One previous study reported lower fluid pH in septic versus nonseptic effusions in cats, but no difference in pH between septic and nonseptic effusions in dogs [12].

The most common causes of pleural effusion in dogs were neoplasia and IVP. This observation is consistent with one previous study of pleural and mediastinal effusions in dogs [14], whereas other reports list CHF and neoplasia as the most common causes of pleural effusion in dogs [4, 13]. Inflammatory pleural effusions frequently were classified as exudates (83%) and had higher degenerate neutrophil counts compared with other disease processes, consistent with the underlying pathophysiology and previous reports [10]. Pleural effusions caused by DOP were uncommon, but when present had lower total protein concentration and NCC than other effusions. This finding is consistent with a previous study that found lower total protein concentrations in nonneoplastic peritoneal or pleural effusions compared with neoplastic effusions in dogs, with this difference being entirely a consequence of the dogs with DOP as the cause of nonneoplastic effusion [11].

Our findings regarding pericardial effusion in dogs are consistent with previous studies, indicating that these effusions are usually hemorrhagic and associated with neoplasia or idiopathic pericarditis [6, 15].

Regarding peritoneal effusion in cats, our findings were similar to a previous report [4] in which FIP was the most common causes of peritoneal effusion in 34 cats. For pleural effusion in cats, the most common causes were IHP and neoplasia, similar to previous retrospective studies indicating that CHF was the most common cause of pleural effusion in cats [16, 17]. Our findings regarding pericardial effusions in cats are consistent with a previous study suggesting that up to 75% of such effusions were associated with CHF [24]. The lack of significant associations between the cause of effusion and cytologic classification in cats is similar to a previous study reporting high-protein transudates and exudates as the most common classifications of cavitary effusions in cats [4] and other studies that found considerable overlap among fluid types and underlying disease for cats with pleural effusion [17–19].

We also identified differences in fluid characteristics among body cavities for a given disease process, a finding not previously reported. Our data suggest that for a given disease process, effusion in the abdomen tends to be more proteinaceous and have higher neutrophil counts compared with effusion in the pleural space in both dogs and cats.

Differences in effusion characteristics between cats and dogs generally reflected the different underlying disease processes common in each species. Peritoneal inflammatory effusions were less cellular and had lower neutrophil counts in cats compared with dogs, likely reflecting the high prevalence of FIP in cats, which causes high-protein mononuclear IVP [16]. A study of dogs and cats with pyothorax found that cats had higher NCC in pleural effusion, with neutrophils more likely to contain intracellular bacteria [10]. A similar trend was not seen in our data that combined different underlying diseases under the disease process category of IVP. Pleural neoplastic effusions had higher lymphocyte counts in cats versus dogs, likely reflecting the high prevalence of lymphoma in the cats in our study. These species-specific differences are important for clinicians to consider when interpreting cytologic findings in cats versus dogs.

Limitations of our study are mainly related to its retrospective design. Diagnoses were based on information available in the medical record and the opinion of the attending clinician or pathologist. There was no standardization of which diagnostic tests were performed for each animal. Dogs and cats were only included if cytologic analysis was performed on at least one effusion, which might introduce sampling bias. Statistical power was limited for comparisons of certain cavities and disease process categories. Cases with cytologic evaluation of pericardial effusion were limited for both dogs and cats.

In conclusion, we report fluid characteristics associated with various causes of cavitary effusions in dogs and cats. Data from our study can provide clinicians with a general range of fluid analytes and cytologic variables expected for different disease process categories in each cavity and species. Cytologic classification and fluid characteristics were most predictive of cause in peritoneal effusions in dogs, whereas in cats, substantial overlap occurred across causes.

Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Authors declare no institutional animal care and use committee or other approval was needed. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Statistical comparisons of fluid characteristics among disease process categories for dogs with fluid analysis performed on peritoneal effusion ($n = 185$) or pleural effusion ($n = 85$). p values are provided for overall group comparisons (when appropriate) and post hoc pairwise testing. DOP, decreased oncotic pressure; IHP, increased hydrostatic pressure; IVP, increased vascular permeability; MN, mononuclear; MP, macrophage; NCC, nucleated cell count; N:M ratio, ratio of neutrophils to mononuclear cells. **Figure S1:** Percentage of effusions caused by various underlying causes classified as transudate, modified transudate, exudate, or hemorrhage. Numbers within each bar indicate absolute numbers of animals with each cytologic classification for each cause; see Tables 2–6 for detailed breakdown.